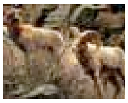

Why do females live longer?
A study investigates why females tend to live longer than males, not just when it comes to humans. **https://dgit.in/apr20-07**

Get your steps in!
A new study found that a higher daily step count leads to all-around lower mortality risk. **https://dgit.in/apr20-08**

Rheology of Cats
Physics, 2017

A paper titled the "Rheology of Cats" was published by MA Fardin in a journal called Rheology Bulletin, which tracks important research in the domain. The purpose of the study was to answer the question, "are cats liquid?", and turns out, the answer is that cats can behave both like liquids and solids. When they occupy a smaller container, they flex and "flow" to fill in all the available space, similar to a liquid, but when placed in a large area such as a bathtub, they stretch out and occupy as much space as possible, behaving more like solids. Note that the discipline of rheology has some interpretations specific to the field that are not really the common versions. For example, irrespective of the state of matter of the cat, it is considered to be liquid based on something known as the "relaxation time", that is how long a cat takes to occupy a container, and mould itself to that shape.

Looking at things upside down
Perception, 2016

A team of researchers were awarded an Ig Nobel prize for a paper published in 2016, titled "perceived size and perceived distance of targets viewed from between the legs: Evidence for proprioceptive theory". The paper was published in 2006, and the prize awarded only 10 years later, but better late than never. The researchers kept objects at varying distances and checked how test subjects could estimate their size and distance while looking at them straight, and while looking at them upside down and between their legs. The results showed that the judgement of size disrupted when they were looked at upside down, and that the nearer objects appeared farther away. Through the use of spectacles with prisms, the researchers were able to demonstrate something important, that the perception had to do with the orientation of the body, and not the inverted image on the retina.

What flies like
Physics, 2016

The prize was awarded to a pair of bizarre studies, with Gábor Horváth and Gyoergy Kriska in both teams of researchers. The first study showed that dragonflies were attracted to black gravestones in a particular graveyard in Hungary. The researchers found that the black stone had reflective properties that were close to that of water, and that the dragonflies behaved like they would around pools of water. The team also investigated what kinds of horses most attracted horseflies, and found that horseflies were most likely to infect brown and black horses. White horses had an advantage, as they were the least attractive to the parasites. The discrepancy was because of how light interacted with the colour and texture of the horse. While white horses more easily get cancer because of sensitivity to ultraviolet light, and suffer from a higher predation risk in the wild, they have a natural defense against horseflies.

Dung beetle navigation
Biology/Astronomy, 2013

While it has long been known that birds, seals and humans can navigate through the night using the stars, such an ability was never before seen in insects. A team of researchers found that dung beetles, in fact, use the Milky Way to guide their movements. Dung beetles were observed in the wild and tested inside a planetarium. When the skies were clear, the dung beetles would roll their dung balls in a straight line, but lose their way when the skies were overcast. In the controlled environment of a planetarium, the dung beetles were able to navigate on the basis of the projected stars. The researchers also turned off particular objects, and found that the beetles were able to navigate even when there was no moon, but lost their way when there was no Milky Way. This was the first time that any animal was conclusively shown to navigate using the galaxy.